

Relational Database Modeling

Project Name: Procurement System

Team Number: 3

Live Working Project (Test Environment): <https://procurement-system-peach.vercel.app/> (<https://procurement-system-peach.vercel.app/>)

1. Business Rules and Scope

System Context

This system models a real-world enterprise procurement platform designed to manage the entire lifecycle of purchasing goods from external vendors. It seamlessly oversees employee requests, manager approvals, purchase order formulation by procurement officers, order fulfillment by vendors, and final invoice and payment processing by the finance department.

Business Rules

1. A **User** is assigned exactly one **Role** and belongs to exactly one **Department**.
2. A **Department** can employ/contain multiple **Users**.
3. An employee (**Users**) can submit multiple **Purchase Requests**.
4. Each **Purchase Request** must belong to a single **Department**.
5. A **Manager** (from **Users**) can review, approve, or reject multiple **Purchase Requests**.
6. A **Purchase Request** can have multiple review records (**PR_Approvals**) and status changes (**PR_Status_History**).
7. A **Procurement Officer** generates **Purchase Orders** derived from approved **Purchase Requests**.
8. A **Purchase Order** must be tied to exactly one **Vendor**.
9. A **Vendor** can fulfill multiple **Purchase Orders** and establish multiple **Contracts**.
10. A **Vendor** can have multiple **Vendor Categories** associated with them.
11. A **Purchase Order** can contain multiple **Line Items**.
12. Upon shipping, a **Purchase Order** generates a **Goods Receipt**, which is logged into the system.
13. The Finance department processes **Invoices** which correspond to shipped **Purchase Orders**.
14. An **Invoice** belongs to one **Vendor**.
15. An **Invoice** can have multiple **Payments** associated with it.
16. **Budget Transactions** are recorded against a Department whenever a **Purchase Order** deducts from the department's budget.
17. Any significant action performed by a User triggers an entry in the **Audit Log**.

Assumptions

- A Vendor interacts directly with the system through their respective Vendor portal account.
- All purchases operate strictly under budget constraints defined at the Department level.
- Single-currency transactions are assumed as the standard.

Out-of-Scope Items

- Bank or credit card payment gateway integration (payments are manually recorded for simulation).
- Detailed HR operations and management logic beyond basic department and role assignment.
- Post-procurement inventory tracking and advanced warehouse management capabilities.

2. Relational Database Model

Logical Schema Design

The relational database utilizes a comprehensively normalized schema to minimize data redundancy and prevent partial or transitive dependencies (satisfying Third Normal Form - 3NF).

- **Authentication & Authorization:** Handled via **Users**, **Roles**, and **Departments**.
- **Procurement Workflow:** Anchored by **Purchase_Requests**, tracked dynamically through **PR_Approvals** and **PR_Status_History**.
- **Vendor & Order Fulfillments:** Centralized by **Vendors**, **Vendor_Categories**, and **Contracts** which feed logically into **Purchase_Orders**, **PO_Line_Items**, and **Goods_Receipt**.
- **Finance:** Administered strictly by **Invoices**, **Payments**, and **Budget_Transactions**.
- **System Traceability:** Accounted for dynamically by **Audit_Log** and **Notifications**.

Entities & Relationships

- **Users (1) to (N) Purchase_Requests**
- **Departments (1) to (N) Purchase_Requests**

- Purchase_Requests (1) to (N) PR_Approvals
- Purchase_Requests (1) to (N) PR_Status_History
- Purchase_Requests (1) to (N) Purchase_Orders
- Vendors (1) to (N) Purchase_Orders
- Vendors (1) to (N) Invoices
- Purchase_Orders (1) to (N) PO_Line_Items
- Purchase_Orders (1) to (1) Goods_Receipt
- Purchase_Orders (1) to (N) Invoices
- Invoices (1) to (N) Payments

3. Data Dictionary

A definitive breakdown of core entities, attributes, and constraints deployed within the relational system

Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Roles	role_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	role_name	TEXT	-	NOT NULL	UNIQUE
	permissions_json	TEXT	-	NULL	-
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Departments	dept_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	dept_name	TEXT	-	NOT NULL	-
	budget_allocated	REAL	-	NULL	DEFAULT 0
	budget_used	REAL	-	NULL	DEFAULT 0
	manager_id	INTEGER	FK	NULL	REFERENCES Users(user_id)
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Users	user_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	name	TEXT	-	NOT NULL	-
	email	TEXT	-	NOT NULL	UNIQUE
	role	INTEGER	FK	NULL	REFERENCES Roles(role_id)
	department_id	INTEGER	FK	NULL	REFERENCES Departments(dept_id)
	created_at	TIMESTAMP	-	NULL	DEFAULT CURRENT_TIMESTAMP
	is_active	BOOLEAN	-	NULL	DEFAULT 1
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Purchase_Requests	pr_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	employee_id	INTEGER	FK	NULL	REFERENCES Users(user_id)
	dept_id	INTEGER	FK	NULL	REFERENCES Departments(dept_id)
	item_name	TEXT	-	NOT NULL	-
	description	TEXT	-	NULL	-
	quantity	INTEGER	-	NOT NULL	-
	estimated_cost	REAL	-	NULL	-
	status	TEXT	-	NULL	DEFAULT 'PENDING'
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
PR_Approvals	approval_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	pr_id	INTEGER	FK	NULL	REFERENCES Purchase_Requests(pr_id)
	manager_id	INTEGER	FK	NULL	REFERENCES Users(user_id)
	decision	TEXT	-	NULL	-
	comments	TEXT	-	NULL	-
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Vendors	vendor_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	company_name	TEXT	-	NOT NULL	-
	contact_name	TEXT	-	NULL	-
	email	TEXT	-	NULL	-
	rating	INTEGER	-	NULL	-
	is_active	BOOLEAN	-	NULL	DEFAULT 1
Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Purchase_Orders	po_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	pr_id	INTEGER	FK	NULL	REFERENCES Purchase_Requests(pr_id)
	vendor_id	INTEGER	FK	NULL	REFERENCES Vendors(vendor_id)
	procurement_officer_id	INTEGER	FK	NULL	REFERENCES Users(user_id)
	delivery_due_date	TEXT	-	NULL	-

	total_amount	REAL	-	NULL	-
	status	TEXT	-	NULL	DEFAULT 'ISSUED'

Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Invoices	invoice_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	po_id	INTEGER	FK	NULL	REFERENCES Purchase_Orders(po_id)
	vendor_id	INTEGER	FK	NULL	REFERENCES Vendors(vendor_id)
	due_date	TEXT	-	NULL	-
	amount	REAL	-	NULL	-
	status	TEXT	-	NULL	DEFAULT 'PENDING'

Table	Column Name	Data Type	Key Form	Null Constraints	Default Rules & Foreign Checks
Payments	payment_id	INTEGER	PK	NOT NULL	AUTOINCREMENT
	invoice_id	INTEGER	FK	NULL	REFERENCES Invoices(invoice_id)
	paid_by	INTEGER	FK	NULL	REFERENCES Users(user_id)
	amount_paid	REAL	-	NULL	-
	payment_method	TEXT	-	NULL	-

(Peripheral system tables tracking historical operations—such as PR_Status_History, Vendor_Categories, PO_Line_Items, Goods_Receipt, Budget_Transactions, Audit_Log, Notifications—implicitly adhere to identical structural compliance bindings regarding standard foreign keys and nullable directives).

5. Additional Tables Data Dictionary

Table	Column Name	Data Type	Key Form	Null Constraints	Description
PR_Status_History	history_id	INTEGER	PK	NOT NULL	AUTOINCREMENT Tracks status changes of purchase requests over time
	pr_id	INTEGER	FK	NOT NULL	References Purchase_Requests(pr_id)
	status	TEXT	-	NOT NULL	Current status
	changed_at	TIMESTAMP	-	NOT NULL	Timestamp of change
Vendor_Categories	vc_id	INTEGER	PK	NOT NULL	AUTOINCREMENT Links vendors to categories (M:N)
	vendor_id	INTEGER	FK	NOT NULL	References Vendors(vendor_id)
	category_name	TEXT	-	NOT NULL	Category name
Contracts	contract_id	INTEGER	PK	NOT NULL	AUTOINCREMENT Formal agreements with vendors
	vendor_id	INTEGER	FK	NOT NULL	References Vendors(vendor_id)
	start_date	DATE	-	NOT NULL	Contract start
	end_date	DATE	-	NULL	Contract end
PO_Line_Items	line_item_id	INTEGER	PK	NOT NULL	AUTOINCREMENT Individual items within a purchase order
	po_id	INTEGER	FK	NOT NULL	References Purchase_Orders(po_id)
	product_name	TEXT	-	NOT NULL	Name of product
	quantity	INTEGER	-	NOT NULL	Quantity ordered
	unit_price	REAL	-	NOT NULL	Price per unit
Budget_Transactions	bt_id	INTEGER	PK	NOT NULL	AUTOINCREMENT Records budget usage per department
	dept_id	INTEGER	FK	NOT NULL	References Departments(dept_id)
	amount	REAL	-	NOT NULL	Transaction amount
	transaction_date	TIMESTAMP	-	NOT NULL	Date of transaction
Audit_Log	log_id	INTEGER	PK	NOT NULL	AUTOINCREMENT System audit entries for actions
	user_id	INTEGER	FK	NOT NULL	References Users(user_id)
	action	TEXT	-	NOT NULL	Description of action
	timestamp	TIMESTAMP	-	NOT NULL	When action occurred
Notifications	notification_id	INTEGER	PK	NOT NULL	AUTOINCREMENT User notifications
	user_id	INTEGER	FK	NOT NULL	References Users(user_id)
	message	TEXT	-	NOT NULL	Notification content
	is_read	BOOLEAN	-	NOT NULL	Read status
	created_at	TIMESTAMP	-	NOT NULL	Creation time

4. Entity-Relationship Diagram (ERD)

```
erDiagram
    Departments ||--o{ Users : "employs"
    Roles ||--o{ Users : "assigns status to"
    Users ||--o{ Purchase_Requests : "initiates"
    Departments ||--o{ Purchase_Requests : "owns & funds"

    Purchase_Requests ||--o{ PR_Approvals : "is reviewed via"
    Purchase_Requests ||--o{ Purchase_Orders : "formulates"
    Purchase_Requests ||--o{ PR_Status_History : "status changes"

    Vendors ||--o{ Purchase_Orders : "fulfills"
    Vendors ||--o{ Vendor_Categories : "has categories"
    Vendors ||--o{ Contracts : "has contracts"

    Purchase_Orders ||--o{ PO_Line_Items : "contains line items"
    Purchase_Orders ||--o{ Goods_Receipt : "generates upon ship"
    Purchase_Orders ||--o{ Invoices : "billed via"

    Invoices ||--o{ Payments : "settled via"

    Departments ||--o{ Budget_Transactions : "records budget usage"
    Users ||--o{ Audit_Log : "creates audit entries"
    Users ||--o{ Notifications : "receives notifications"

    Departments {
        int dept_id PK
        text dept_name
        real budget_allocated
    }
    Users {
        int user_id PK
        text name
        text email
        int role FK
        int department_id FK
    }
    Roles {
        int role_id PK
        text role_name
    }
    Purchase_Requests {
        int pr_id PK
        int employee_id FK
        int dept_id FK
        text status
        real estimated_cost
    }
    PR_Approvals {
        int approval_id PK
        int pr_id FK
        int manager_id FK
        text decision
    }
    PR_Status_History {
        int history_id PK
        int pr_id FK
        text status
        timestamp changed_at
    }
    Vendors {
        int vendor_id PK
        text company_name
    }
    Vendor_Categories {
        int vc_id PK
        int vendor_id FK
    }
```

```

        text category_name
    }
    Contracts {
        int contract_id PK
        int vendor_id FK
        date start_date
        date end_date
    }
    Purchase_Orders {
        int po_id PK
        int pr_id FK
        int vendor_id FK
        real total_amount
        text status
    }
    PO_Line_Items {
        int line_item_id PK
        int po_id FK
        text product_name
        int quantity
        real unit_price
    }
    Goods_Receipt {
        int receipt_id PK
        int po_id FK
        date received_date
    }
    Invoices {
        int invoice_id PK
        int po_id FK
        int vendor_id FK
        real amount
        text status
    }
    Payments {
        int payment_id PK
        int invoice_id FK
        int paid_by FK
        real amount_paid
        text payment_method
    }
    Budget_Transactions {
        int bt_id PK
        int dept_id FK
        real amount
        timestamp transaction_date
    }
    Audit_Log {
        int log_id PK
        int user_id FK
        text action
        timestamp timestamp
    }
    Notifications {
        int notification_id PK
        int user_id FK
        text message
        boolean is_read
        timestamp created_at
    }
}

```